

Levels of AI Autonomy and Analyst Control for Bias Mitigation Interventions in Visual Analytics

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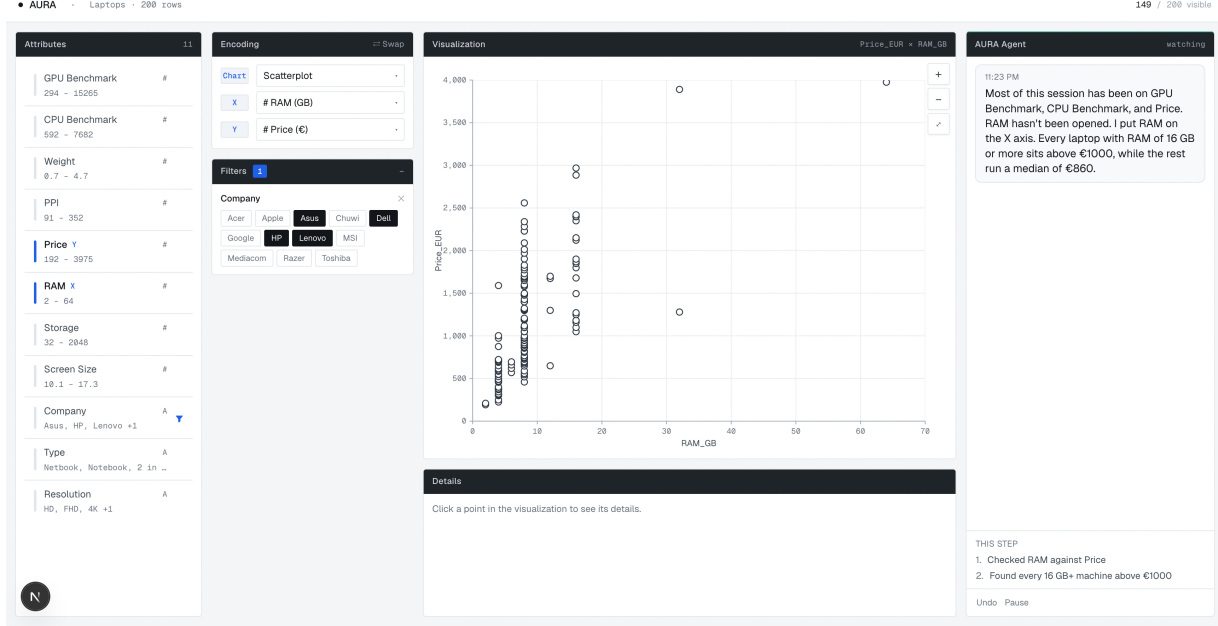


Figure 1: AURA in the *Agentic* condition. After the analyst focuses on GPU, CPU, and Price, the agent brings unexamined RAM into view and reports a finding (right-hand side). *Socratic* and *Directive* conditions share the panel and trigger but leave view changes to the analyst: *Socratic* asks a question, prompting reflection, while *Directive* specifies what to explore next.

ABSTRACT

Visual analytics tools help analysts explore datasets effectively with the power of human perception. However, analysts’ exploration is often uneven, informed by their preferences, experiences, and potential biases. As a result, some data points and attributes are visited repeatedly while others remain underexplored. To address this uneven exploration, we implemented a system that measures bias in the analyst’s *process* from their interactions in real time and triggers an AI intervention. We implemented three levels of autonomy. *Socratic AI* prompts reflection by asking the analyst questions to help them reason and act. *Directive AI* explicitly names underexplored attributes and tells the analyst to examine them. *Agentic AI* changes the underexplored view directly and reports what is there. Since the three conditions share the same detector and trigger, only the degree of the LLM agent’s intervention at each step varies. This system provides a basis for studying how analyst control and reflection may change as AI autonomy increases.

Index Terms: Visual Analytics, Cognitive Bias, Agentic AI.

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1 INTRODUCTION

Visual analysis is a powerful way for analysts to explore and understand data intuitively by leveraging visual perception. Tools like Tableau and Power BI allow analysts to turn structured datasets into visual representations, helping people digest data more easily. However, the freedom of visual exploration and representation can be a double-edged sword: because analysts’ attention is seldom distributed uniformly, they may leave some parts over- or under-examined.

Nonuniform exploration can reflect task goals, expertise, or cognitive biases such as confirmation bias [27]. Wall et al. showed that interactions form a record of the analyst’s thought process and proposed metrics to measure how a person samples the data space during analysis [27]. In this paper, we refer to skew in that sampling as *process* bias. We use the term descriptively rather than to imply harm, since nonuniform sampling may reflect preferences, experience, and expertise that is not inherently harmful. Because these metrics are applied to interaction sequences, they can report skew while the analyst is still working, opening the possibility of real-time intervention. The Lumos system by Narechania and colleagues took this step, using in-situ and ex-situ interaction traces to give real-time feedback on analytic behavior [17]. However, increasing awareness of exploration patterns does not always change behavior. The Lumos study found many moments of awareness, where participants compared their traces against a baseline and reflected on the difference, but temporal analysis of their interactions provided little evidence that the traces reduced potentially biased

analytical behavior [17].

Mitigating potential biases for many individuals requires more than awareness. Prior work argues that debiasing information should be both informative and actionable [30], and awareness alone leaves the burden of finding the appropriate responsive action to the user. Related, prior work characterizes guidance degrees as orienting, directing, and prescribing [4]. Building on this progression, we describe mitigation of process bias as a three-step loop: noticing that attention has narrowed, reasoning about what to explore next, and acting by changing the view.

AI systems can now facilitate those steps at different levels of autonomy. An agent can prompt the analyst to reflect on their exploration, suggest what to explore next, or change the view itself. Dhanoa et al. describe agentic visualization as systems in which agents take on analytical work while the analyst retains authority over the interpretation [7]. A process-bias intervention can fit this framing by assigning different parts of the mitigation loop to the agent.

To make this tension explicit, we present AURA: Analytical Understanding through Reflective Asking, a system that detects process bias in real time during visual data analysis and delivers an LLM intervention at one of the three levels of AI autonomy: *Socratic*, *Directive*, or *Agentic*. We refer to these levels as the **delegation ladder**. Our contribution is thus the design and implementation of three levels of delegation for process-bias interventions. AURA provides a controlled setting for investigating the effects of different levels of guidance on process bias, perceived control, and metacognitive awareness, which remain open empirical questions. Because the levels share the same detector, trigger, interface, and limit of three interventions per session, future evaluations can examine these relationships while varying which parts of the mitigation loop are delegated to the agent.

2 BACKGROUND AND RELATED WORK

2.1 Measuring and Mitigating Interaction Bias

Interactions with a visual analytics tool leave a meaningful record of the analyst’s behavior. Dou et al. found that human coders could reconstruct much of an analyst’s reasoning from interaction logs [8], and Brown et al. predicted personality traits and task performance from the same signal [1]. Subsequent work has formalized these logs to predict clicks, quantify exploration breadth, and track analytic focus [20, 10, 32].

These interaction-log methods characterized what analysts were doing but did not indicate whether the exploration was skewed [8, 1, 20, 10, 32]. Addressing this gap, Wall et al. proposed six metrics that assess how a user samples the data space against a baseline of unbiased analysis, modeled as a Markov chain over the space of possible interactions [27]. Which baseline to compare against remains open [26]. We adopt these metrics in our system because our intervention needs a signal that the exploration has narrowed in real time. We use the term *process bias* for what these metrics capture, a property of the exploration itself rather than of the conclusion the analyst reaches [28].

2.2 Reflection in Visual Analytics

Reflection during analysis means turning attention from the data to the process of examining it, asking not only what the data shows but whether the way it is being examined is sound. Metacognition, the monitoring and regulation of one’s own reasoning, has been studied extensively in learning and cognitive science but has entered visualization only recently [11, 19]. A systematic review of the concept in visualization searched a decade of publications across six visualization venues and five metacognition venues, and found twenty-one papers at the intersection, reporting that metacognition is rarely an explicit focus [5]. The same absence has been noted across

human-computer interaction, where metacognition has been considered mainly within computer science education, even as generative systems place growing metacognitive demands on their users [23].

What exists instead are adjacent constructs that stand in for metacognition without naming it. Bias-aware interfaces quantify an analyst’s exploration against a statistical baseline and surface the result during analysis [27, 17, 29]. Sensemaking accounts describe how analysts construct and question the frames through which they read a display, activities that are metacognitive in substance [5]. In each case the analyst is informed about their own process, and what follows is left to them.

Lumos supported awareness of exploration patterns, yet temporal analysis provided little evidence of reduced potentially biased behavior [17]. More broadly, visualization studies rarely give analysts an opportunity to see and revise their strategies [5]. Ptolemy’s Bearings feature monitors exploration within a semantic map and flags large semantic jumps to prompt reflection [31]. Such feedback leaves analysts to decide how to respond. AURA varies how much of this response is delegated to the agent across its *Socratic*, *Directive*, and *Agentic* conditions.

2.3 Agents in Visual Analytics

Early work on agents in visual analytics concentrated on translating a stated analytic request into an appropriate view [22, 18]. Recent systems assume considerably more of the analytic loop. A multi-modal language model coupled to a visualization pipeline allows an agent to evaluate the view it has produced and revise the encoding accordingly [15], and subsequent work extends this to direct control of the tool’s own interface [14]. Other systems decide which relationships in the data merit examination [16] and assemble the resulting views into a complete report [12]. This growth has prompted efforts to codify recurring agent-based design patterns [7].

An agent can therefore carry out the step that the systems in the preceding section withhold. Detecting a skew in an analyst’s own exploration is well established. How much of the response to delegate remains an open question. Proactive systems offer data-focused recommendations, ranking candidate views by statistical interest or weighing the benefits of unsolicited assistance against its interruption cost [24, 6, 13]. The agentic systems above generally begin with an analyst’s request. Feng et al. describe how responsibility shifts between users and agents through roles ranging from operator to observer [9]. AURA focuses on this allocation within individual process-bias interventions.

What such delegation would cost the analyst is largely unmeasured. Agentic systems are assessed primarily on their output, through case studies, expert feedback, and the quality of the views they generate [15, 16, 12]; what the analyst retains of the exploration is seldom reported. Designs that most reduce overreliance are also the least liked [3], and overreliance can decrease when verification becomes easier [25]. Sustained AI assistance has been linked to skill degradation in expert practice [2]. Whether greater agent autonomy necessarily reduces analyst control and reflection during exploratory analysis remains open.

3 THE DELEGATION LADDER

Noticing and mitigating process bias requires three things to happen. The analyst has to notice that attention has narrowed, determine what to explore next, and act on it. In the metamemory framework, the first is monitoring, information flowing from the object level up to the meta level, and the third is control, the meta level acting on what the object level does next [19].

Parasuraman et al.’s model distinguishes four types of automation: information acquisition, information analysis, decision and action selection, and action implementation [21]. Building on this

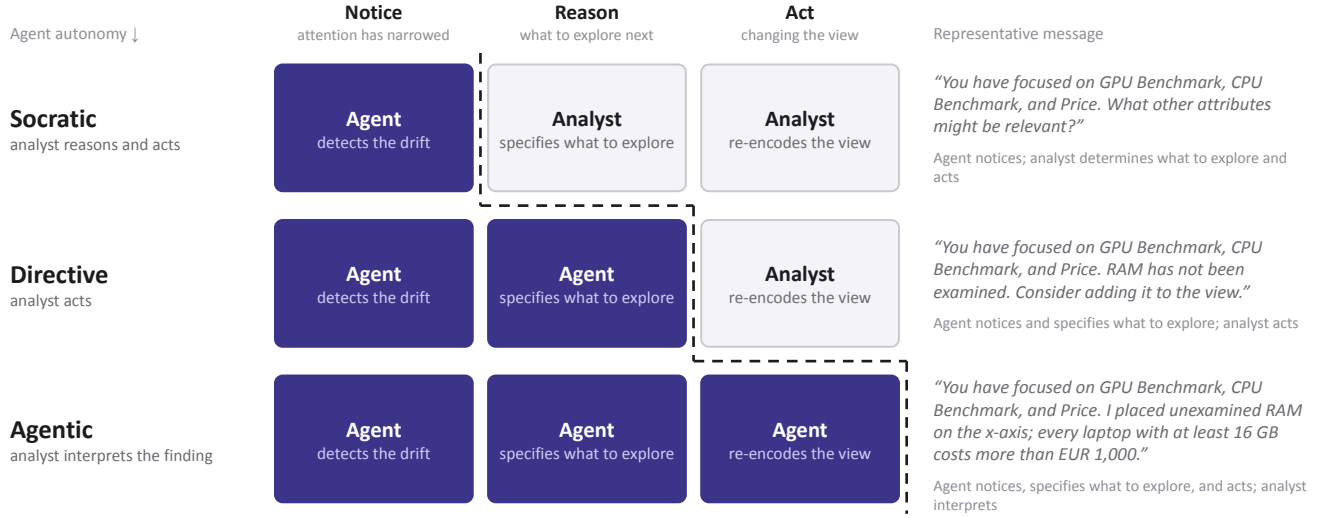


Figure 2: The delegation ladder within an individual intervention. Each stage performs the same work across all three conditions; only ownership shifts. The dashed boundary encloses what the analyst still performs, and it recedes by one stage at each level. Moving down the ladder delegates more of the mitigation loop to the agent, leaving fewer opportunities for the analyst to monitor and regulate their exploration.

research, our delegation ladder specifies how responsibility is allocated within a process-bias intervention. We distinguish reporting the observed exploration pattern (Notice), telling the analyst what to explore next (Reason), and changing the view accordingly (Act). Notice leaves the next exploration target to the analyst; Reason specifies it. The three levels depicted in Fig. 2 progressively delegate these responsibilities to the agent, providing a basis for studying analyst control and reflection.

Socratic AI takes the first step alone. It notices that attention has narrowed and asks the analyst about it, without naming what has been left out. The analyst still has to determine which attribute to explore next and whether to bring it into view by assigning a new axis attribute or adjusting a filter. *Directive AI* takes the first two steps. It notices the skew and tells the analyst which attribute to explore next. The analyst is told where to look but still must choose to look, so the action remains theirs. *Agentic AI* takes all three steps. It notices the skew, determines which attribute to explore next, changes the encoding or filter itself, and reports what the view contains. The agent stops there. What the finding means for the task, however, is left to the analyst, who is now reading a view they did not build.

The three levels progressively automate one intervention loop while sharing the detector, intervention rules, and panel. This creates a conceptual tension between automation and reflection: delegating more steps may broaden exploration while the analyst performs less monitoring and control. The ladder captures responsibility within individual interventions, but does not imply a corresponding ordering of influence over the broader analytical direction. Even *Socratic* questions may shape what analysts explore next.

4 SYSTEM

AURA is a web-based visual analytics tool for conducting exploratory data analysis (Fig. 1, Fig. 3). The intervention aims to prompt reflection on visualization exploration patterns and help analysts assess whether their exploration aligns with their intentions. Analysts explore a tabular dataset through a scatterplot, changing axes, applying filters, and inspecting each individual data point. Every interaction is logged with a timestamp, its target (where applicable), and whether the analyst or the agent performed it.

From the log, the detector computes four of Wall et al.’s interac-

tion bias metrics [27]. Data point coverage (DPC) and attribute coverage (AC) ask how much of the data, or of an attribute’s range, the analyst has reached relative to a baseline in which every data point is equally likely to be examined. Data point distribution (DPD) asks whether individual points were revisited evenly, and attribute distribution (AD) asks whether values touched match those in the full dataset. Because none of the four registers concentration across the set of available attributes, we add a fifth signal, *attribute concentration*, that compares attribute-visit frequencies (i.e., axis changes, filter interactions, and attribute-panel clicks) against a uniform baseline.

The interventions, described in the following paragraphs, are triggered when any of the five scores remains above its corresponding threshold for at least 45 seconds within a 60-second window. All five thresholds default to 0.5 and are configurable. After each intervention, the system uses a 240-second cooldown, and at most three interventions per session serve as the cap. These parameters are provisional design settings shared across all three conditions. The sustained trigger filters out brief threshold crossings, while the cooldown spaces interventions apart, and the cap limits their total number. Future studies should calibrate the thresholds and timing parameters for the intended users and tasks.

Socratic AI receives as input the attributes the analyst has concentrated on and returns as output a question about that pattern. The question names what has drawn attention but not what has been left out, so it points at the analyst’s behavior rather than at the data. Nothing on the chart changes. An example intervention is:

You have focused most of this session on GPU Benchmark, CPU Benchmark, and Price. What other attributes might be relevant to your analysis?

Directive AI receives the same description and tells the analyst to explore the least-examined attribute next. The chart still does not change. For the same exploration state:

You have focused most of this session on GPU Benchmark, CPU Benchmark, and Price. RAM has not been examined. Consider adding it to the view.

Agentic AI receives the same description and selects an action from a closed set of eight using function calling, all of which the

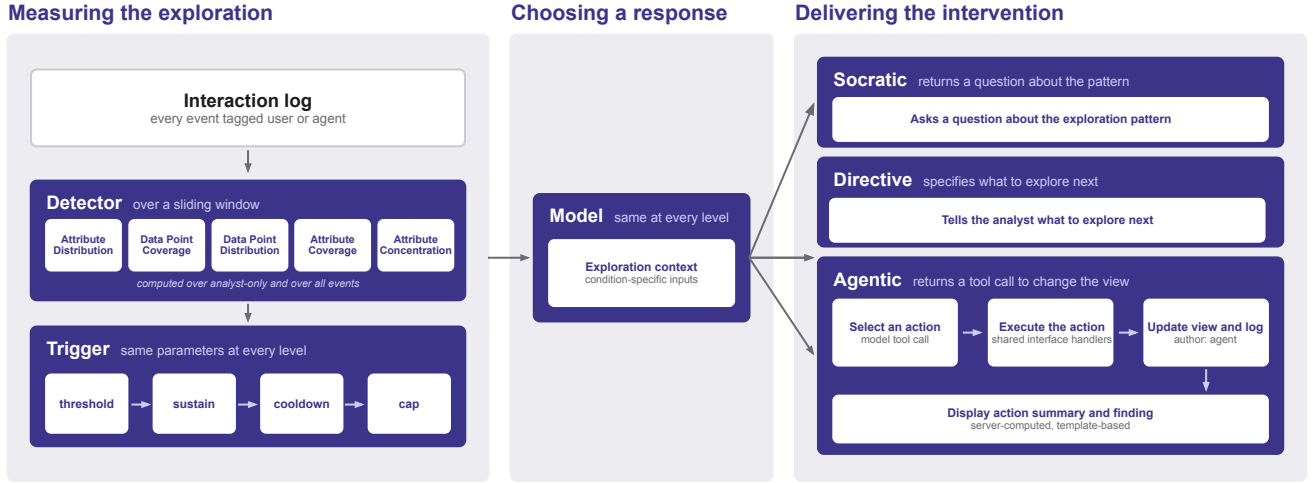


Figure 3: AURA’s pipeline. All conditions share the detector and trigger rules. *Socratic* and *Directive* display messages, while *Agentic* also updates the chart, logs the action, and reports a computed finding.

analyst can also perform: setting either axis, changing the chart type or aggregation, selecting an aggregate group, setting a numeric range, toggling a category, adding a filter, and selecting a point. The system executes it through the same handler the interface calls. The analyst can undo the action or pause the agent for the remainder of the session. The agent’s action enters the same log the detector reads, so each event records its author. For the same exploration state:

You have focused most of this session on GPU Benchmark, CPU Benchmark, and Price. RAM has not been examined, so I placed it on the x-axis. Every laptop with at least 16 GB of RAM costs more than EUR 1,000, while the others have a median price of EUR 860.

5 DISCUSSION

5.1 The Tension Between Automation and Reflection

Agentic visualization defines the agent as independently performing analytical tasks while the analyst guides the process, interprets results, and maintains decision-making authority [7]. The division assumes the delegated work produces something the analysis needs.

However, AURA’s intervention primarily redirects attention to information already available in the dataset. The neglected attribute is already in the dataset and already reachable in the interface. What has to happen is that the analyst turns toward it. The agent can bring the attribute into view, but this action does not ensure that the analyst noticed the original skew or reconsidered their exploration strategy. Successful action and successful reflection can therefore diverge.

Parasuraman et al. warned that automation does not simply take over human activity but changes it [21]. Monitoring and control are relations inside the analyst’s cognition [19]. An agent can notice that attention has narrowed without the analyst noticing it. It can bring the neglected attribute into view without the analyst having decided to look for it. What arrives is a fact about the data. What may be skipped is the analyst’s reflection on why that view was needed. This divergence may improve the current view without helping the analyst recognize or regulate the same pattern in later exploration.

5.2 Evaluating Analyst Control and Reflection

Evaluating AURA requires examining both exploration coverage and the analyst’s control and reflection. This distinction is difficult

to evaluate because a more capable agent may produce a more complete view. Yet every step the agent takes can improve what the analyst sees while removing an occasion for them to look. Autonomy is therefore a design parameter whose effects are not fully captured by the resulting view. Agentic systems are assessed on what they produce, through case studies, expert feedback, and the quality of the views they generate [15, 16, 12]. An evaluation based only on exploration coverage could rate a session in which the agent opened every neglected view as thorough, even if the analyst initiated none of these changes.

The cost is also hard to notice from inside the session. At the *Agentic* level, analysts retain responsibility for interpretation while the agent performs the intervention. We hypothesize that analysts may still feel in control even if they monitor their exploration less actively. Future evaluation should therefore examine perceived control alongside evidence of monitoring and reflection.

5.3 Future Work

Future work can compare the three levels against a no-intervention baseline to examine how autonomy affects process bias, perceived control, and metacognitive awareness. These outcomes should be considered together, since lower process bias may coincide with reduced control or reflection. AURA also records whether the analyst undoes or pauses the agent, so a reported loss of control can be read against what they did about it.

6 CONCLUSION

We presented AURA, a visual analytics system that organizes process-bias interventions into *Socratic*, *Directive*, and *Agentic* levels. At the *Socratic* level, the agent notices the skew, while the analyst determines what to explore next and changes the view. At the *Directive* level, the agent also specifies what to explore next, leaving the analyst to act. At the *Agentic* level, the agent completes the loop by changing the view itself. This design highlights a central tension: greater autonomy may broaden exploration while reducing opportunities for analysts to reflect on and regulate their own process. AURA provides a basis for studying how process bias, perceived control, and metacognitive awareness change as more of this loop is delegated.

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